

UNIVERSITY OF BAHRAIN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

SECOND SEMESTER 2023 -2024

TEST 2 (A) - Key

MATHS. 211

DATE: 8 - 5 - 2024

TIME: 11:00 PM - 12:00 PM

EXAM INSTRUCTIONS:

- 1) Number of pages is 5 including the cover page.
 - 2) Show all your work, except for question one.
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NAME :	I.D. NUMBER:	SECTION:	SERIAL NUMBER:

<i>Q.1</i>	<i>Q.2</i>	<i>Q.3</i>	<i>Q.4</i>	<i>OUT OF</i>
<i>16</i>	<i>10</i>	<i>6</i>	<i>8</i>	

GOOD LUCK

QUESTION 1 (2 marks for each question)

Fill in the blanks to make a correct statement:

1) Suppose that $(p)_B = (1, 2, -2)$ where $B = \{1 + x, 1 - x^2, x - x^2\}$ is a basis of P_2 .

then

$$p = 3 - x$$

2) Suppose that $S = \{u_1, u_2, u_3, u_4, u_5\}$ spans $\mathbb{R}^n$. Is $n > 5$ or $n \leq 5$?

$$n \leq 5$$

3) Let $S = \{2, 4\sin^2(x), \cos^2(x)\}$. Is S linearly dependent or linearly independent?

linearly dependent

4) If $A' = \begin{bmatrix} 1 & -2 & 2 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ is a row-echelon form matrix of A , then a basis of the row space of A is

$$\{(1, -2, 2, 0), (0, 1, 2, 3), (0, 0, 1, 0)\}$$

5) If A is a matrix of size 6×4 , then the largest possible value of $\text{rank}(A)$ is equal to

$$4$$

6) If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ 1 & 3 & 3 \end{bmatrix}$, then the rank of A is equal to

$$3$$

7) If A is a matrix of size 6×5 and $\text{nullity}(A^T) = 2$, then

$$\text{nullity}(A) = 1$$

8) If $\{1 + X + X^2, 2 + X, a + X\}$ is a basis of P_2 , then

$$a \neq 2$$

QUESTION 2 (4 + 4 + 2 marks)

(Show your work)

a) Show that $B = \{u, v, w\}$ is a basis of $\mathbb{R}^3$, where

$$u = (2, 1, 0) \quad , \quad v = (1, 0, 1) \quad , \quad w = (0, -1, 1)$$

Solution: Since B has 3 elements and $\dim \mathbb{R}^3 = 3$, it is sufficient to prove that B is linearly independent.

$$au + bv + cw = 0 \quad \Rightarrow \quad a(2, 1, 0) + b(1, 0, 1) + c(0, -1, 1) = (0, 0, 0) \quad \mathbf{2}$$

$$\Rightarrow \quad 2a + b = 0 \quad \mathbf{1}$$

$$a - c = 0$$

$$b + c = 0$$

$$\Rightarrow \quad a = b = c = 0 \quad \mathbf{1}$$

b) Find the coordinates of $h = (1, 2, 3)$ relative to B .

Solution:

$$au + bv + cw = h \quad \Rightarrow \quad a(2, 1, 0) + b(1, 0, 1) + c(0, -1, 1) = (1, 2, 3) \quad \mathbf{1}$$

$$\Rightarrow \quad 2a + b = 1$$

$$a - c = 2$$

$$b + c = 3$$

$$\Rightarrow \quad a = -4, b = 9, c = -6 \quad \mathbf{2}$$

$$\Rightarrow \quad (h)_B = (-4, 9, -6) \quad \mathbf{1}$$

c) Without any calculations, determine whether $\{u, v, w, h\}$ is linearly independent.

Solution:

The set $\{p_1, p_2, p_3, p\}$ is linearly dependent because it consists of 4 vectors in the vector space $\mathbb{R}^3$ with dimension 3. **2**

QUESTION 3 (6 marks)

Find the dimension of $V = \text{Span}\{f_1, f_2, f_3\}$, where $f_1 = x$, $f_2 = e^x$, $f_3 = e^{-x}$

Solution:

The set $\{f_1, f_2, f_3\}$ is a spanning set of V . To prove that it is a basis, it suffices to show that it is linearly independent, equivalently to prove that $W(x)$ is not the zero function. **1**

We have

$$\begin{aligned} W(x) &= \begin{vmatrix} x & e^x & e^{-x} \\ 1 & e^x & -e^{-x} \\ 0 & e^x & e^{-x} \end{vmatrix} & \mathbf{1} \\ &= e^x e^{-x} \begin{vmatrix} x & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} & \mathbf{1} \end{aligned}$$

So

$$W(x) = \begin{vmatrix} x & 0 & 0 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} = 2x \quad \mathbf{1}$$

That is,

$$W(x) = 2x \neq \text{zero function.} \quad \mathbf{1}$$

Hence, $\dim V = 3$ **1**

QUESTION 4 (4 + 4 marks)

Let $A' = \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$ be a row-echelon form of A .

a) Find a basis of the nullspace of A .

Solution:

Let us transform the matrix A' to a reduced row-echelon form

$$A' \sim \begin{bmatrix} \mathbf{1} & 2 & -1 & 0 & -1 \\ 0 & \mathbf{0} & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 1 \end{bmatrix} \sim \begin{bmatrix} \mathbf{1} & 2 & 0 & 0 & -1 \\ 0 & \mathbf{0} & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 1 \end{bmatrix} = A'' \quad \mathbf{1}$$

$$\begin{bmatrix} \mathbf{1} & 2 & 0 & 0 & -1 \\ 0 & \mathbf{0} & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 1 \end{bmatrix} X = O \Rightarrow x_1 + 2x_2 - x_5 = 0 ; x_3 = 0 ; x_4 + x_5 = 0 \quad \mathbf{1}$$

Let $x_2 = r$, $x_5 = s$ be free variables, then the leading variables are $x_1 = -2r + s$, $x_3 = 0$ and $x_4 = -s$.

Thus $N(A) = \{ (-2r + s, r, 0, -s, s) : r, s \text{ are real numbers} \} = \text{Span}\{(-2, 1, 0, 0, 0), (1, 0, 0, -1, 1)\}$. $\mathbf{1}$

Since $\dim N(A) = \text{Nullity}(A) = 5 - 3 = 2$, then $\{(-2, 1, 0, 0, 0), (1, 0, 0, -1, 1)\}$ is a basis of $N(A)$. $\mathbf{1}$

b) If c_1, c_2, c_3, c_4, c_5 are the column vectors of A , write c_2 and c_5 as linear combination of c_1, c_3, c_4 .

Solution:

Let $c_1', c_2', c_3', c_4', c_5'$ be the column vectors of A'' .

Then $c_2' = 2c_1' + 0c_3' + 0c_4'$ and $c_5' = -1c_1' + 0c_3' + 1c_4'$. $\mathbf{2}$

It follows that $c_2 = 2c_1 + 0c_3 + 0c_4$ and $c_5 = -1c_1 + 0c_3 + 1c_4$. $\mathbf{2}$